robust stock market combined with early death—a very low probability. Overall, the chance that a 55-year-old man who retires at age 45 with only $200,000 will be able to support a spending habit of $16,000 per year is not great.

Before I go further, I must emphasize the assumptions that go into such a Monte Carlo simulation study. First, I made a moderate assumption about human longevity patterns. If you believe that you are healthier than the average American, then your probabilities of sustainability or success are even lower than the preceding estimates. Remember, if you are healthier than average, then a different mortality table (representing a selected healthy population group) might apply to you. In this case you might be more likely to live much longer and consume longer. All else being equal, this reduces the chances of sustainability.

Second, I assumed in the simulation that the real, after-inflation rate of return from equity markets would be 7% with a volatility of 20%. These numbers correspond with the behavior of American equity markets during the last fifty years. Please note: I am not assuming that you will earn 7% in real terms each year. Rather, I'm assuming that in the long term, you'll earn an average of 7% per annum, with a volatility of 20%. (Remember that volatility is a measure of how wide the spectrum of investment returns is expected to be. Again, a volatility number of 20% means that 95% of the time, the returns will be within $2 \times 20\% = 40\%$ of the expected value.) Admittedly, 7% may be a bit aggressive; a variety of financial commentators believe that the equity risk premium, as it is called, will be much less than observed in the past.

Do I Need a Supercomputer?

No, you don't. In fact, one of the achievements I'm most proud of is a one-line formula that I developed and then published a few years ago with some colleagues of mine that eliminates the need for cumbersome and expensive retirement income computer simulations in many (although not all) cases. Yes, the hunting trophy on my mantle is a mathematical relationship that arrives at the probability of retirement sustainability, given a spending and an investing strategy. To